

TRAFFIC ANALYTICS SYSTEM

(Vehicle Detection, Tracking, Direction Analysis, and Speed Estimation from Fixed-Camera Video)

1. INTRODUCTION

Traffic monitoring and analysis play a crucial role in intelligent transportation systems, urban planning, and road safety.

This project presents an end-to-end traffic analytics system that processes a fixed-camera video to automatically detect vehicles, track them across frames, determine their direction of travel, estimate their speed, and generate structured analytical outputs. The system is designed with a modular, object-oriented architecture and focuses on robustness, scalability, and clarity of results.

The primary objectives of this project are:

- To accurately detect and track multiple vehicle types
- To determine direction of movement (IN / OUT)
- To estimate vehicle speed from video data
- To generate both visual and structured outputs for analysis

2. SYSTEM OVERVIEW AND ARCHITECTURE

The system follows a pipeline-based architecture, where each component is responsible for a single well-defined task. All logic is implemented using an object-oriented approach, avoiding global variables and ensuring maintainability.

High-level workflow:

- Read frames from a fixed-camera input video
- Detect vehicles in each frame
- Track detected vehicles across frames
- Assign unique IDs and vehicle classes
- Analyze direction of movement
- Estimate vehicle speed using pixel-to-meter conversion with perspective handling
- Log structured results and generate visual outputs

Each stage operates independently but communicates through well-defined data structures, allowing easy modification or extension of individual components.

3. DATASET AND INPUT CHARACTERISTICS

The system is designed for fixed-camera videos, typically captured from:

- Gate cameras
- Overhead or semi top-down road cameras
- Stationary surveillance cameras

Input characteristics

- Single static camera (no camera motion)
- Mixed vehicle types
- Variations in:
 - Lighting conditions
 - Traffic density
 - Partial occlusions
 - Vehicle sizes and speeds

The system does not rely on any pre-labeled dataset and instead operates directly on raw video input, making it suitable for real-world deployments.

4. APPROACH AND METHODOLOGY

This section describes the technical approach used to implement each component of the traffic analytics pipeline.

4.1 Vehicle Detection

Vehicle detection is performed using YOLOv8 (Ultralytics), a state-of-the-art real-time object detection model. YOLOv8 provides:

- High detection accuracy
- Support for multiple vehicle classes
- Efficient inference suitable for video processing

Only relevant COCO classes are mapped to project-specific vehicle categories:

- Car
- Bike / Two-wheeler
- Bus
- Truck

4.2 Multi-Object Tracking

To maintain consistent identities across frames, the system uses BoT-SORT, a modern multi-object tracking algorithm that combines:

- Motion prediction
- Appearance-based association
- Robust ID assignment

Each detected vehicle is assigned a unique track ID, which remains stable throughout its visible lifetime in the video. This enables accurate counting, direction analysis, and speed estimation.

4.3 Vehicle Counting

Vehicle counting is performed based on unique track IDs:

- Each vehicle is counted only once

- Counts are maintained per vehicle category
- A total vehicle count is computed dynamically

Live per-class and total vehicle counts are overlaid on the annotated output video for real-time visualization.

4.4 Direction Analysis

Since the camera is fixed and top-down in nature, direction is inferred using net vertical displacement of tracked vehicle centroids:

- Vehicles moving towards the bottom of the frame → IN
- Vehicles moving towards the top of the frame → OUT

Direction is finalized only after sufficient movement is observed to avoid noisy or ambiguous classifications. Once determined, the direction remains fixed for that vehicle.

4.5 Speed Estimation

Speed estimation is performed using a pixel-to-meter conversion approach with additional perspective handling.

Key aspects

- Recent trajectory points are used instead of the full track
- Small pixel movements are ignored as noise
- Perspective scaling is applied based on vertical position in the frame
- Speed is computed in km/h
- Physically unrealistic speeds are clamped to a safe upper bound

To improve robustness:

- Instantaneous speeds are accumulated
- Final speed per vehicle is computed as the average moving speed

This approach provides stable and realistic speed estimates even in the presence of temporary stops or occlusions.

5. OUTPUT AND RESULTS

5.1 Annotated Output Video

The system generates an annotated video containing:

- Thin bounding boxes around vehicles
- Vehicle ID and class
- Direction (IN / OUT)
- Estimated speed
- Live per-class counts
- Total vehicle count

This visual output allows quick qualitative evaluation of system performance.

5.2 Structured CSV Outputs

Per-Vehicle Results (results.csv)

Each row corresponds to a unique vehicle and includes:

- Serial number
- Track ID
- Vehicle type
- Direction of travel
- Average moving speed (km/h)
- Start frame
- End frame

Speed Summary (speed_summary.csv)

This file contains:

- Average speed per vehicle category
- Speed distribution across predefined ranges
- Vehicle counts per category
- Total vehicle count

Separating raw results from summary statistics keeps the data clean and easy to analyze.

6. DEPLOYMENT

6.1 Command-Line Deployment

The primary deployment mode is a local command-line application.
The pipeline can be executed using:

```
python -m src.pipeline
```

This mode is suitable for batch processing and reproducible analysis.

6.2 Web-Based Demo Deployment

To demonstrate the system as an interactive application, a lightweight web-based demo was implemented using Streamlit.

Features of the web demo

- Browser-based interface
- Upload any .mp4 traffic video
- One-click execution of the analytics pipeline
- Outputs generated locally

The web demo runs entirely on the local machine and does not require cloud deployment, making it easy to test and demonstrate.

7. EVALUATION

The system was evaluated qualitatively based on:

- Detection accuracy across vehicle types
- Stability of tracking IDs
- Correctness of direction classification
- Realism and consistency of speed estimates
- Correctness of per-class and total counts

The annotated video and CSV outputs provide clear evidence of correct system behavior across varying traffic conditions.

8. CHALLENGES AND FAILURE CASES

Challenges

Several technical challenges were encountered during development:

- Speed estimation without camera calibration, leading to reliance on approximate pixel-to-meter conversion
- Maintaining consistent vehicle IDs in dense traffic and occlusion scenarios
- Class ambiguity between visually similar vehicles (e.g., car vs small truck)
- Accurate vehicle counting, requiring careful handling of unique IDs to prevent double counting
- Direction estimation during temporary stops, where negligible centroid movement caused ambiguity
- Detecting small objects, particularly two-wheelers, in top-down camera views

Failure Cases

- Heavily occluded vehicles may experience delayed tracking
- Extreme camera angles may reduce speed accuracy
- Very dense traffic can increase processing time

9. IMPROVEMENTS AND REFINEMENTS

Several improvements were implemented during development:

- Bounding boxes and background overlays were refined to reduce visual clutter in dense scenes
- Vehicle counting was made event-driven, ensuring each vehicle is counted only once
- Direction was locked after first stable detection to avoid fluctuations
- Speed estimation was smoothed using moving averages and noise thresholds
- BoT-SORT replaced centroid-only tracking to reduce ID switching
- YOLOv8 model variants and confidence thresholds were adjusted to improve two-wheeler detection

10. LIMITATIONS AND FUTURE IMPROVEMENTS

Current Limitations

- CPU-based inference results in slower processing for long videos
- Pixel-to-meter conversion relies on approximate calibration
- Single-camera assumption

Future Improvements

- GPU acceleration using CUDA or TensorRT
- Camera calibration using homography
- Multi-camera support
- Real-time streaming support
- Cloud or edge deployment for scalability

11. CONCLUSION

This project demonstrates a complete and practical traffic analytics system built using modern computer vision techniques. By combining detection, tracking, direction analysis, and speed estimation into a unified pipeline, the system provides meaningful traffic insights from a single fixed-camera video.

The modular design, structured outputs, unit testing, and deployment options make the solution robust, extensible, and suitable for real-world applications as well as further research and development.